Topic : Sonar based Target Ship Detection using Machine Learning.

ABSTRACT

The project is to develop an accurate and reliable system capable of analyzing sonar signals generated from ship wake disturbances and determining the presence or absence of a target ship. The proposed system utilizes advanced signal processing, machine learning techniques to automatically learn the distinguishing characteristics of target and non-target acoustic signatures.

The input to the system consists of underwater sonar signals collected from ship wake regions. These signals are preprocessed to reduce noise and enhance relevant information, followed by transformation into suitable representations that effectively capture the temporal and spectral characteristics of the acoustic data. The processed data are then used to train and evaluate intelligent classification models capable of performing binary classification, where the output indicates whether a target ship is present or absent.

Upon successful detection of a target ship, the system further supports target localization and tracking by estimating the target's movement characteristics and direction. This information can be utilized by an autonomous guidance module to continuously update the trajectory toward the detected target. The overall framework aims to provide accurate target identification, reduced false alarm rates, robust performance in noisy underwater environments, and efficient real-time operation. The expected outcome of this project is a lightweight, reliable, and highly accurate underwater target detection and tracking system suitable for defense-oriented applications.

Keywords: Sonar Signals, Ship Wake Analysis, Underwater Target Detection, Binary Classification, Target Tracking.